Weekly Assignment: Variables, Sets, Relations, and Functions (Epp 1.1, 1.2, & 1.3)

Complete each of the following problems. Show all work and justify your answers.

Problem 1 (Epp 1.1 #2). In each part, fill in the blanks using a variable or variables to rewrite the given statement.

Is there an integer that has a remainder of 2 *when it is divided by* 5 *and a remainder of* 3 *when it is divided by* 6*?*

a. Is there an integer n such that n has ____________?

b. Does there exist ____________ such that if n is divided by 5 the remainder is 2 and if ____________?

Note: There are integers with this property. Can you think of one?

Problem 2 (Epp 1.1 #4). In each part, fill in the blanks using a variable or variables to rewrite the given statement.

Given any real number, there is a real number that is greater.

a. Given any real number r, there is ____________ s such that s is ____________.

b. For any ____________, ____________ such that $s > r$.

Problem 3 (Epp 1.1 #7 b,d). Rewrite the following statements less formally, without using variables. Determine, as best as you can, whether each statement is true or false.

b. There is a real number x such that $x^2 < x$.

d. For all real numbers a and b, $|a + b| \leq |a| + |b|$.

Problem 4 (Epp 1.2 #2 b,d). Write in words how to read each of the following out loud.

b. $\{x \in \mathbf{R} \mid x \leq 0 \text{ or } x \geq 1\}$

d. $\{n \in \mathbf{Z}^+ \mid n \text{ is a factor of } 6\}$

Problem 5 (Epp 1.2 #4). Answer each of the following questions. Give reasons for your answers.

a. Is $2 \in \{2\}$?

b. How many elements are in the set $\{2, 2, 2, 2\}$?

c. How many elements are in the set $\{0, \{0\}\}$?

d. Is $\{0\} \in \big\{\{0\}, \{1\}\big\}$?

e. Is $0 \in \big\{\{0\}, \{1\}\big\}$?

Problem 6 (Epp 1.3 #2). Let $C = D = \{-3, -2, -1, 1, 2, 3\}$ and define a relation S from C to D as follows: For every $(x, y) \in C \times D$,

$$(x, y) \in S \quad \text{means that} \quad \frac{1}{x} - \frac{1}{y} \text{ is an integer.}$$

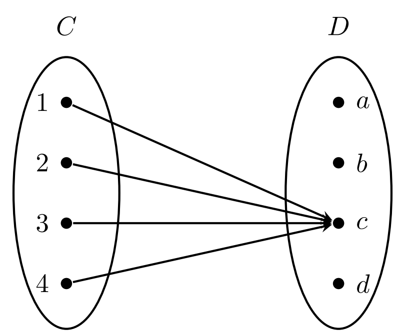
a. Is $2\ S\ 2$? Is $(-1)\ S\ (-1)$? Is $(3, 3) \in S$? Is $(3, -3) \in S$?

b. Write S as a set of ordered pairs.

c. Write the domain and co-domain of S.

d. Draw an arrow diagram for S.

Problem 7 (Epp 1.3 #14). Let $C = \{1, 2, 3, 4\}$ and $D = \{a, b, c, d\}$. Define a function $G\colon C \to D$ by the following arrow diagram:



a. Write the domain and co-domain of G.

b. Find $G(1)$, $G(2)$, $G(3)$, and $G(4)$.

Problem 8 (Proof). Prove that for any equation of the form $y^A + x^B = C$, where A, B, and C are positive integers, whether or not the relation corresponding to this equation is a function (with x as input and y as output, over the reals) depends only upon whether A is even or odd.

Hint: Consider two cases based on the parity of A. For each case, think about how many real values of y satisfy $y^A = C - x^B$ for a given x. When does a real number have a unique real Ath root, and when might it have zero or two?